

MUTUAL FUND ANALYTICS DASHBOARD

Final Capstone Project Report

Submitted By

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Bluestock Mutual Fund Analytics Capstone Project

Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- Streamlit
- Matplotlib

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Bluestock for providing the opportunity to work on this Mutual Fund Analytics Capstone Project. The project offered valuable exposure to real-world financial datasets and enabled me to apply concepts from data engineering, analytics, software development, and business intelligence in a practical environment.

I would also like to acknowledge the guidance and resources that supported the successful completion of this project. Throughout the project, I gained hands-on experience in building ETL pipelines, designing databases, performing exploratory data analysis, implementing financial performance metrics, conducting risk analysis, and developing interactive dashboards.

This project significantly strengthened my understanding of data-driven decision making and financial analytics, while also enhancing my technical skills in Python, SQL, data visualization, and dashboard development.

ABSTRACT

The Mutual Fund Analytics Dashboard project was developed to create an end-to-end analytics platform capable of processing, analyzing, and visualizing mutual fund industry data. The project integrates multiple datasets including NAV history, Assets Under Management (AUM), SIP trends, folio growth, category flows, portfolio holdings, and investor transaction records.

A structured ETL pipeline was implemented to clean and transform raw datasets before storing them within a SQLite database. Exploratory data analysis was performed to identify growth trends, investor participation patterns, and portfolio characteristics. Performance analytics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to evaluate mutual fund performance. Advanced analytics including Value at Risk (VaR), Conditional Value at Risk (CVaR), Investor Cohort Analysis, SIP Continuity Analysis, and Portfolio Concentration Analysis were conducted to assess risk and investor behavior.

The final solution includes an interactive Streamlit dashboard that enables users to explore key insights through visualizations and analytical summaries. The project demonstrates how financial datasets can be transformed into actionable insights using modern data analytics techniques.

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1.INTRODUCTION

The Indian mutual fund industry has witnessed rapid growth over the past decade due to increasing investor awareness, digital investment platforms, and widespread adoption of SIPs. As the volume of available financial data continues to grow, investors and analysts require efficient tools to transform raw information into meaningful insights.

Mutual fund analysis involves evaluating multiple dimensions such as returns, risk, portfolio composition, investor participation, and industry growth. Performing such analysis manually can be both time-consuming and inefficient. This project addresses these challenges by developing a centralized analytics platform that integrates data engineering, financial analytics, and dashboard visualization.

The primary objectives of the project were to build an ETL pipeline, clean and validate mutual fund datasets, design a SQLite database, perform exploratory data analysis, calculate performance and risk metrics, analyze investor behavior, develop a recommendation engine, and create an interactive dashboard for decision support.

The project focuses on historical analysis rather than real-time forecasting. Its scope includes data processing, descriptive analytics, performance evaluation, risk assessment, investor behavior analysis, and dashboard development.

2.DATA SOURCES

Multiple datasets were utilized to provide a comprehensive view of the mutual fund ecosystem.

The Fund Master dataset contained information regarding scheme names, AMFI codes, fund houses, categories, expense ratios, risk levels, and investment requirements. This dataset acted as the primary reference table throughout the project.

The NAV History dataset consisted of daily NAV records for forty mutual fund schemes between January 2022 and May 2026. This dataset formed the basis for return calculations and performance analytics.

The SIP Industry Trends dataset provided monthly SIP inflow figures and was used to evaluate investor participation and industry growth.

The AUM History dataset contained historical Assets Under Management data for leading fund houses and was used to compare market share and industry dominance.

The Folio Growth dataset tracked investor account growth and served as a measure of retail participation in mutual funds.

The Category Flow dataset recorded inflows and outflows across different mutual fund categories and helped identify investor preferences.

The Fund Holdings dataset provided information on portfolio composition, sector allocations, and stock weightages, enabling concentration analysis through HHI calculations.

The Investor Transactions dataset contained more than 32,000 transaction records and was used for investor cohort analysis and SIP continuity assessment.

3.ETL DESIGN AND DATA PROCESSING

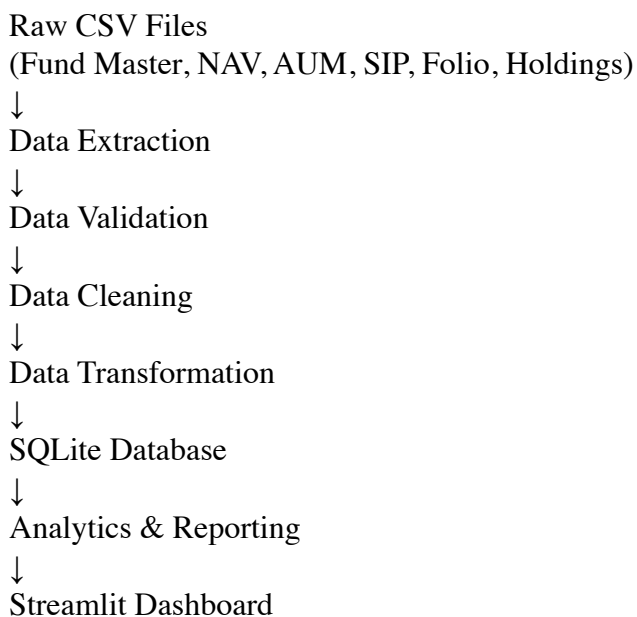
The Extract, Transform, and Load (ETL) process formed the foundation of this project. Since the datasets originated from multiple sources and contained different formats, a structured ETL pipeline was required to ensure consistency, reliability, and accuracy before analysis could be performed.

3.1 Data Extraction

The extraction phase involved importing raw datasets into Python using the Pandas library. The datasets included mutual fund master data, NAV history, AUM records, SIP trends, folio growth information, category-wise fund flows, portfolio holdings, and investor transaction records. Data was collected in CSV and tab-separated file formats and loaded into DataFrames for further processing.

The extraction process ensured that all datasets were successfully loaded and validated before proceeding to the cleaning stage. File structures, column names, and record counts were verified to identify potential issues early in the workflow.

ETL WORKFLOW DIAGRAM



3.2 Data Validation

After extraction, each dataset underwent a validation process to identify data quality issues. Validation checks included:

- Detection of missing values
- Identification of duplicate records
- Verification of date formats
- Validation of numerical fields
- Consistency checks across related datasets

These checks helped ensure that inaccurate or incomplete records would not affect downstream analytical calculations.

3.3 Data Cleaning

Data cleaning involved correcting inconsistencies and preparing the datasets for analysis. The following activities were performed:

- Standardization of date formats
- Removal of duplicate records
- Correction of data type mismatches
- Validation of financial metrics
- Handling of missing values
- Standardization of column naming conventions

Special attention was given to historical NAV data because even minor inconsistencies can significantly impact return calculations and performance metrics.

3.4 Data Transformation

Several new variables were created during the transformation stage to support analytical tasks.

Daily returns were calculated using historical NAV values. Additional metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR were generated from the transformed datasets.

Investor transaction data was also transformed to support cohort analysis and SIP continuity assessment. These transformations enabled deeper analysis of investor behavior and risk characteristics.

3.5 Data Storage

Following cleaning and transformation, the datasets were stored in a SQLite database. Centralized storage improved data organization and allowed efficient retrieval of information for analytical scripts and dashboard development.

The database served as the primary data repository throughout the project and ensured consistency across all analytical modules.

3.6 ETL Outcomes

The ETL pipeline successfully transformed raw mutual fund datasets into structured and analysis-ready data assets. The resulting datasets supported exploratory analysis, performance analytics, risk modeling, investor behavior studies, and dashboard visualization.

4.DATABASE DESIGN

A structured database design was implemented to organize the project datasets efficiently and support analytical processing.

The database was developed using SQLite because it is lightweight, portable, and well suited for analytical applications. A star-schema-inspired structure was adopted to simplify querying and improve data management.

4.1 Database Architecture

The database architecture consisted of one primary dimension table and multiple fact tables.

The dimension table stored descriptive information about mutual fund schemes, while fact tables stored transactional and historical records related to NAV values, AUM, SIP flows, folio growth, category flows, portfolio holdings, and benchmark data.

This design enabled efficient aggregation and simplified analytical calculations.

4.2 Dimension Table

The dim_fund table served as the central reference table within the database.

Key attributes included:

- AMFI Code
- Scheme Name
- Fund House
- Category
- Sub-category
- Expense Ratio
- Risk Category
- Fund Manager
- Benchmark Information

The dimension table allowed other datasets to be linked consistently using AMFI codes.

4.3 Fact Tables

Several fact tables were implemented to store historical and transactional information.

fact_nav

Stored daily NAV records for all analyzed mutual fund schemes. This table formed the foundation for return calculations and performance analytics.

fact_aum

Contained historical Assets Under Management data for major fund houses and supported market share analysis.

fact_sip

Stored monthly SIP inflow information and was used to evaluate investor participation trends.

fact_folio

Contained folio growth statistics and enabled analysis of investor account growth over time.

fact_category_flow

Recorded category-wise inflows and outflows across mutual fund segments.

fact_holdings

Stored portfolio composition information including stock allocations, sectors, and weight percentages.

fact_benchmark

Contained benchmark index data used for Alpha and Beta calculations.

4.4 Database Validation

Validation procedures were conducted after database creation to ensure data integrity.

The validation process included:

- Row count verification
- Primary key consistency checks
- Data type validation
- Successful relationship mapping
- Query testing

The results confirmed that all datasets were successfully loaded and available for analytical processing.

4.5 Benefits of the Database Design

The implemented database structure provided several advantages:

- Centralized data storage
- Faster analytical processing
- Reduced redundancy
- Improved scalability
- Simplified dashboard integration

The database design played a critical role in supporting both analytical workflows and dashboard functionality.

5.EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the mutual fund industry and identify patterns within the datasets before conducting advanced analytics.

The objective of EDA was to uncover trends related to industry growth, investor participation, fund house dominance, portfolio composition, and category preferences.

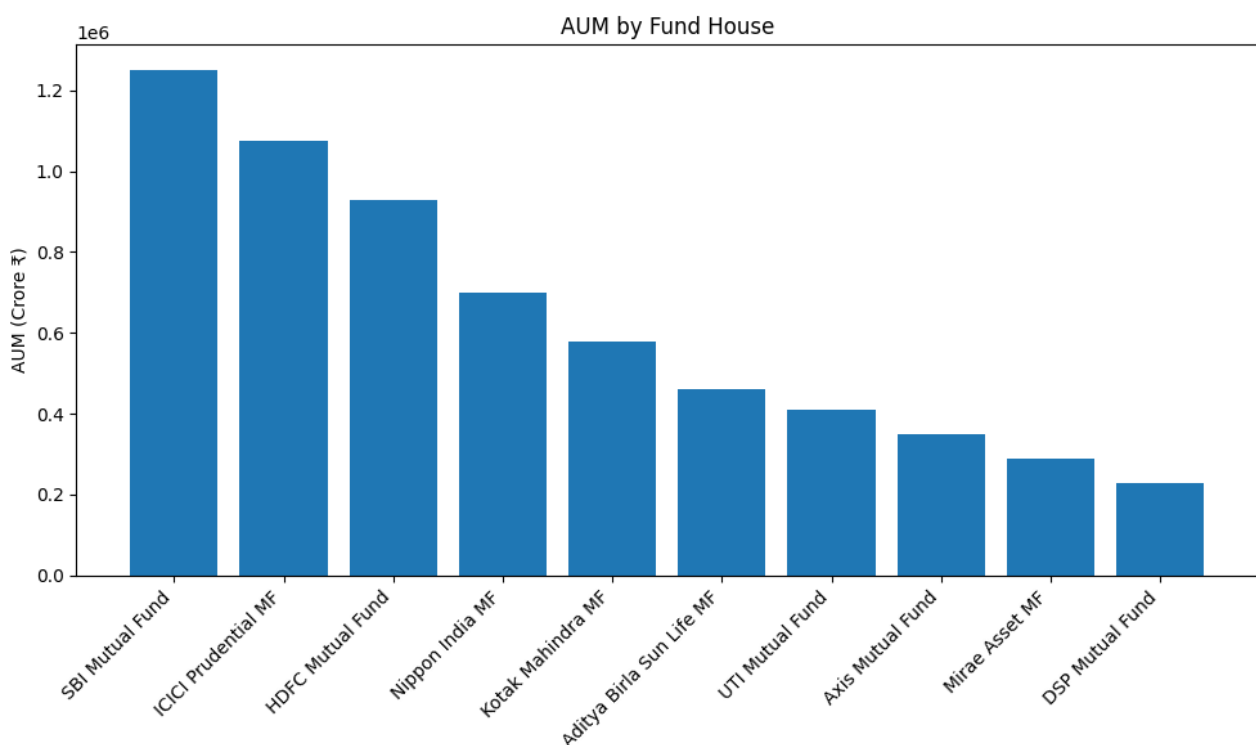
5.1 Assets Under Management (AUM) Analysis

Assets Under Management represent the total market value of assets managed by a mutual fund company and are often used as an indicator of investor trust and market presence.

The analysis revealed a high concentration of assets among a few leading fund houses.

SBI Mutual Fund emerged as the largest asset manager with an AUM exceeding ₹12,50,000 crore. It was followed by ICICI Prudential Mutual Fund and HDFC Mutual Fund.

These results indicate that a small number of established institutions dominate a significant portion of the mutual fund market.



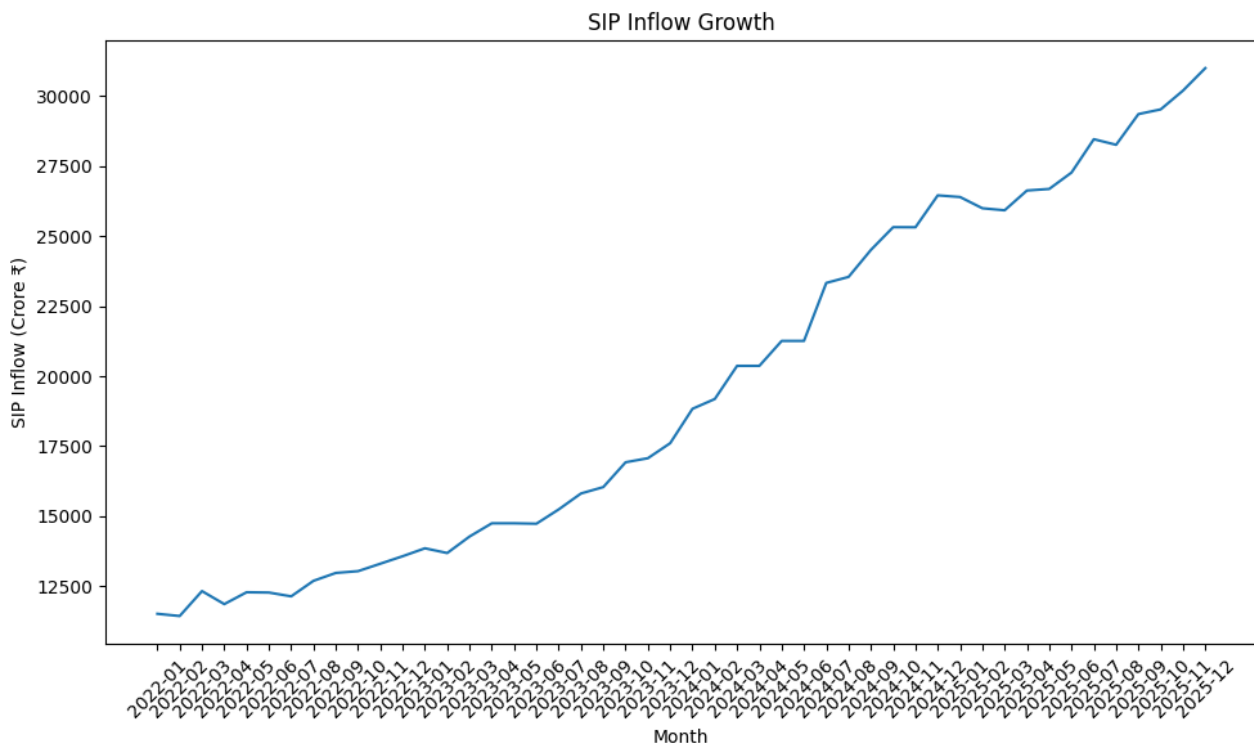
5.2 SIP Growth Analysis

Systematic Investment Plans have become one of the most popular investment methods among retail investors due to their simplicity and long-term wealth creation potential.

The analysis showed significant growth in monthly SIP inflows throughout the study period.

Monthly SIP contributions increased from approximately ₹11,517 crore to ₹31,002 crore, representing growth of more than 169%.

This trend highlights increasing investor confidence and growing participation in mutual fund investments.

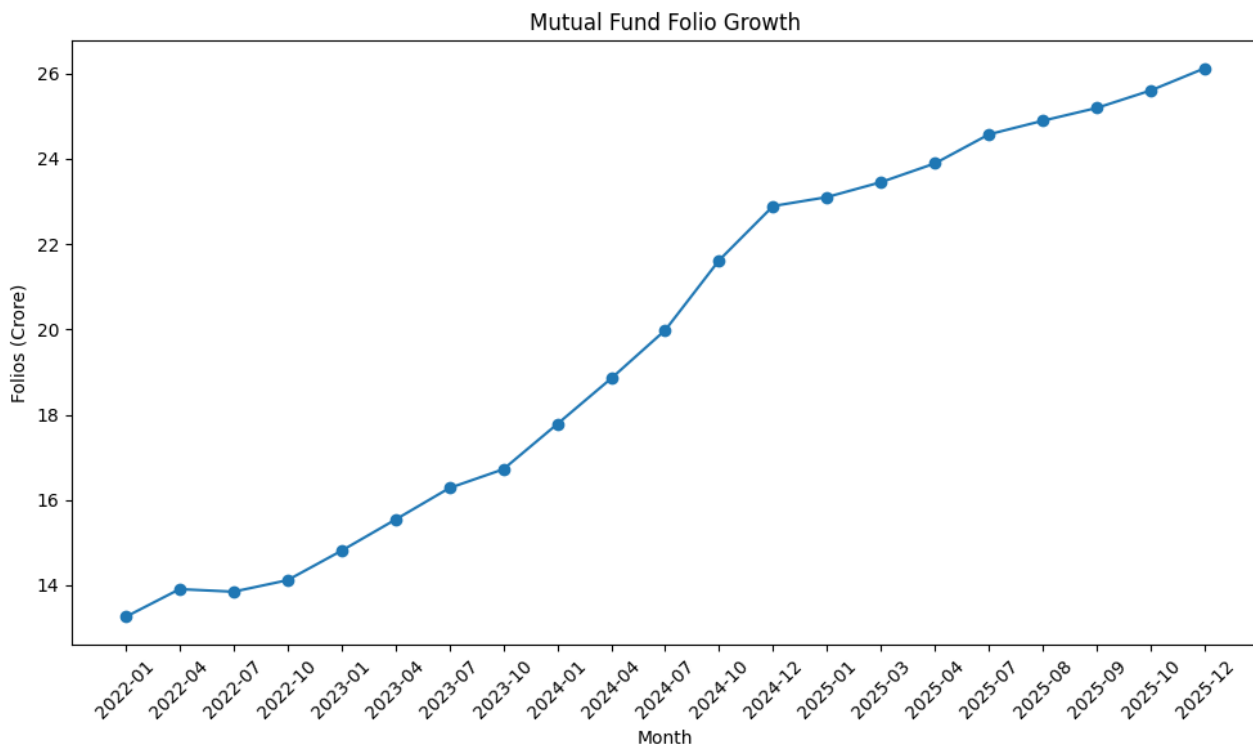


5.3 Folio Growth Analysis

Investor folios provide an indication of market participation and the number of active investment accounts.

The folio analysis demonstrated consistent growth over time, with total folios reaching approximately 26.12 crore by the end of the analysis period.

The results suggest continued expansion of retail investor participation in the mutual fund industry.

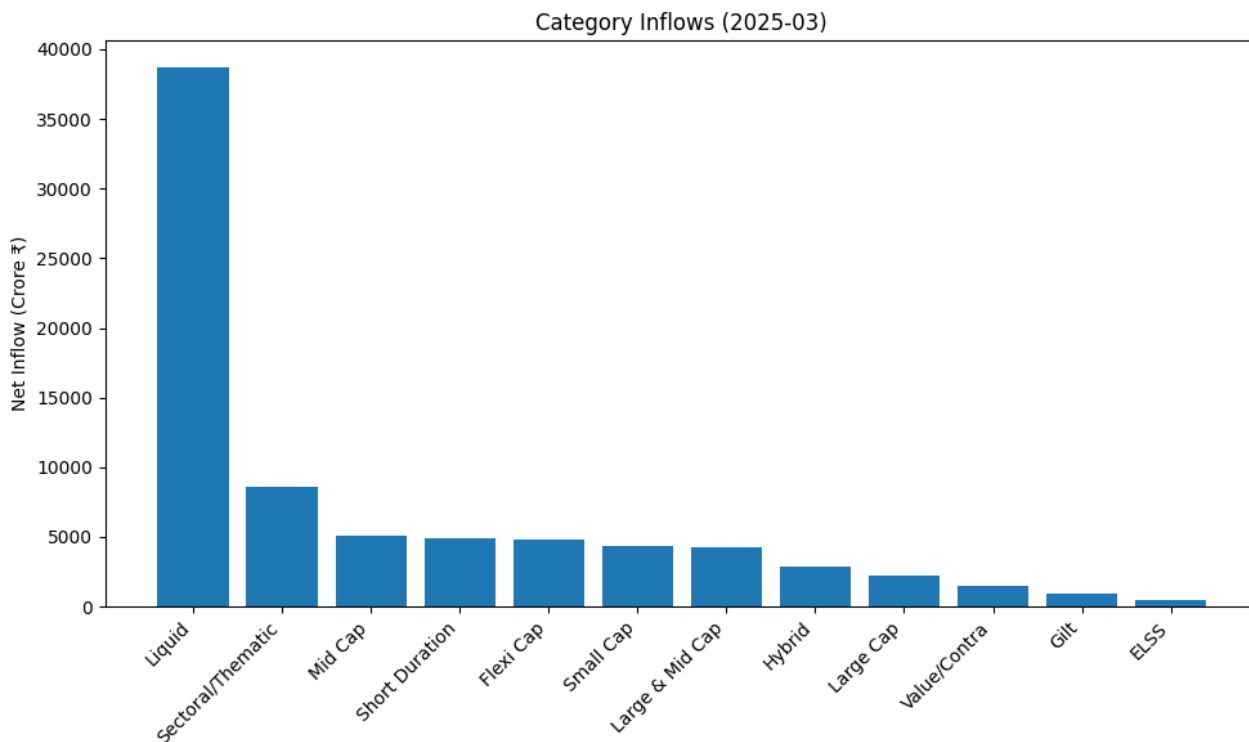


5.4 Category Flow Analysis

Category flow analysis was conducted to evaluate investor preferences across different mutual fund segments.

The results revealed varying levels of inflows and outflows among categories, reflecting changing investor sentiment and market conditions.

Equity-oriented schemes generally attracted stronger inflows during favorable market periods, while debt-oriented schemes displayed more conservative investment patterns.



5.5 Portfolio Holdings Analysis

The holdings dataset was analyzed to identify dominant sectors across mutual fund portfolios.

Banking emerged as the most represented sector, followed by Information Technology, Pharmaceuticals, Automobiles, Utilities, and Infrastructure.

This distribution highlights the importance of these sectors within diversified mutual fund portfolios and reflects broader trends within the Indian economy.

5.6 Key Findings from EDA

The exploratory analysis generated several important insights:

- SBI Mutual Fund maintained the highest AUM among analyzed fund houses.
- SIP inflows increased by more than 169% during the study period.
- Total investor folios exceeded 26 crore.
- Banking and Information Technology were the most represented sectors.
- Investor participation continued to expand steadily across multiple mutual fund categories.

These findings provided the foundation for subsequent performance and risk analytics performed during later stages of the project.

6.PERFORMANCE ANALYTICS

Performance analytics were conducted using historical NAV data for 40 mutual fund schemes. The objective was to evaluate the return-generating capability of each scheme while accounting for risk and market behavior. Various quantitative metrics were calculated to provide a comprehensive assessment of fund performance.

6.1 Daily Return Analysis

Daily returns were calculated using the following formula:

$$\text{Daily Return} = (\text{NAV}_t / \text{NAV}_{t-1}) - 1$$

This metric measures the percentage change in NAV from one trading day to the next and forms the basis for all subsequent performance calculations.

Analysis of the daily return distribution revealed:

- Mean Daily Return: 0.063%
- Standard Deviation: 1.03%
- Minimum Daily Return: -5.81%
- Maximum Daily Return: 6.47%

The distribution appeared reasonable for equity-oriented mutual funds and reflected expected market volatility over the study period.

6.2 CAGR Analysis

Compound Annual Growth Rate (CAGR) was used to evaluate long-term fund performance. CAGR provides a standardized measure of annualized returns and enables fair comparison between different schemes.

The analysis was performed for one-year and three-year investment horizons.

Among the strongest performers were:

- Axis Midcap Fund
- Mirae Asset Large Cap Fund
- ICICI Prudential Bluechip Fund
- HDFC Mid-Cap Opportunities Fund
- SBI Bluechip Fund

These funds demonstrated strong annualized growth while maintaining competitive risk-adjusted performance.

6.3 Sharpe Ratio Analysis

The Sharpe Ratio evaluates returns generated per unit of risk.

The formula used was:

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma \times \sqrt{252}$$

Where:

R_p = Portfolio Return

R_f = Risk-Free Rate (6.5%)

σ = Portfolio Volatility

The analysis identified several funds with exceptional risk-adjusted performance.

Top Sharpe Ratio Funds:

1. Mirae Asset Large Cap Fund (1.45)
2. Kotak Flexicap Fund (1.31)
3. Mirae Asset Tax Saver Fund (1.23)
4. SBI Bluechip Fund (1.21)
5. ICICI Prudential Midcap Fund (1.18)

A higher Sharpe Ratio indicates more efficient conversion of risk into returns.

6.4 Sortino Ratio Analysis

While the Sharpe Ratio penalizes all volatility, the Sortino Ratio focuses exclusively on downside volatility.

This metric provides a more investor-focused evaluation because investors are generally concerned with losses rather than positive fluctuations.

Funds with strong Sortino Ratios demonstrated superior downside risk management and more consistent return generation.

6.5 Alpha and Beta Analysis

Alpha and Beta were calculated using linear regression against benchmark returns.

Alpha measures excess returns generated beyond what would be expected from market exposure alone.

Beta measures sensitivity to market movements.

Interpretation:

- Alpha > 0 indicates benchmark outperformance.
- Alpha < 0 indicates underperformance.
- Beta > 1 indicates higher volatility than the market.
- Beta < 1 indicates lower volatility than the market.

Several schemes generated positive Alpha values, suggesting successful active management.

6.6 Maximum Drawdown Analysis

Maximum Drawdown measures the largest decline from a portfolio peak to its subsequent trough.

This metric is important because it helps investors understand potential capital loss during adverse market conditions.

Funds with lower drawdowns demonstrated stronger resilience during periods of market stress and volatility.

6.7 Fund Scorecard

To provide a comprehensive ranking framework, a composite Fund Scorecard was developed.

Weights used:

- 30% Three-Year CAGR Rank
- 25% Sharpe Ratio Rank
- 20% Alpha Rank
- 15% Expense Ratio Rank
- 10% Maximum Drawdown Rank

The resulting scorecard offered a balanced assessment of return potential, risk-adjusted performance, cost efficiency, and downside protection.

6.8 Key Findings from Performance Analytics

The major findings from this stage were:

- Mirae Asset Large Cap Fund achieved the highest Sharpe Ratio.
- Several mid-cap schemes generated strong CAGR values.
- Positive Alpha values were observed among multiple actively managed funds.
- Risk-adjusted metrics provided more reliable fund rankings than return metrics alone.
- Maximum Drawdown analysis highlighted meaningful differences in downside protection across schemes.

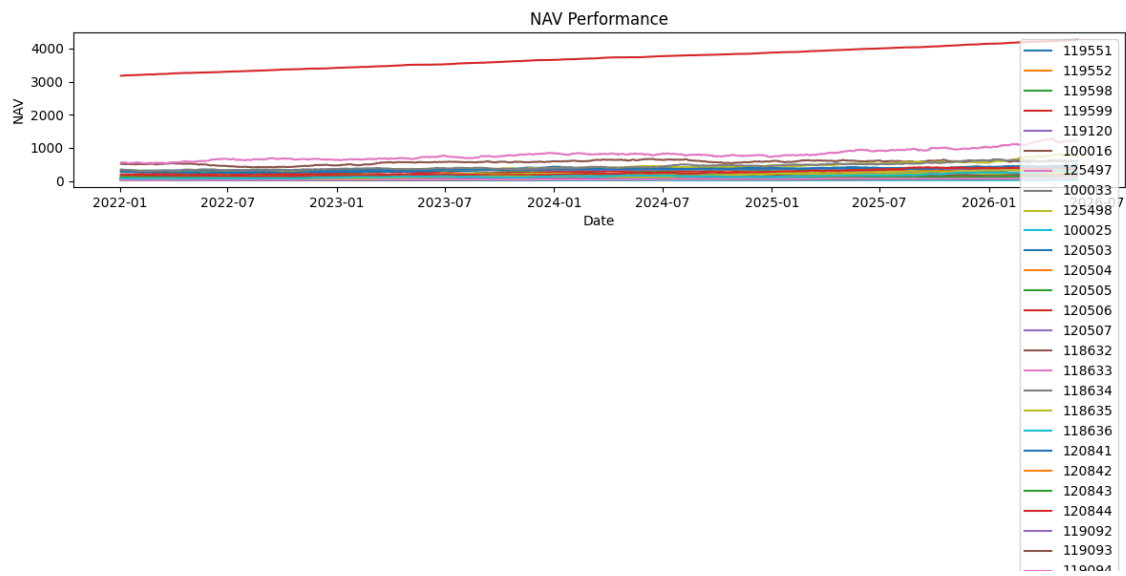


Figure 5: NAV Performance Comparison Across Mutual Funds

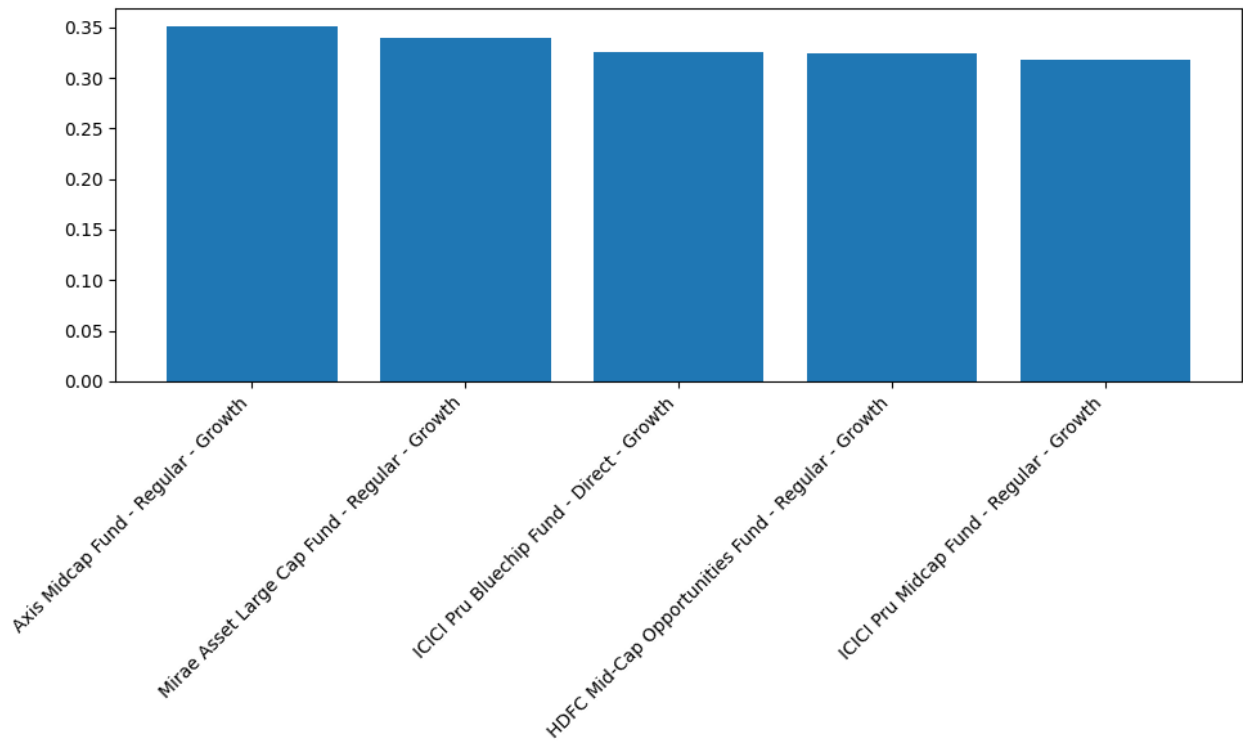


Figure 6: Top CAGR Performing Funds

7 ADVANCED ANALYTICS

While traditional performance metrics focus on returns and volatility, advanced analytics were implemented to gain deeper insights into risk exposure, investor behavior, portfolio concentration, and fund suitability. These techniques extend beyond basic financial analysis and provide a more comprehensive understanding of mutual fund performance.

7.1 Value at Risk (VaR)

Value at Risk (VaR) is one of the most widely used risk management metrics in the financial industry. It estimates the maximum expected loss over a specified period at a given confidence level.

For this project, Historical VaR was calculated at a 95% confidence level using daily return distributions.

The results indicated that certain schemes exhibited significantly higher downside risk than others.

Funds with the highest downside risk included:

- AMFI Code 119599
- AMFI Code 119095
- AMFI Code 101207
- AMFI Code 118634
- AMFI Code 119598

These funds recorded larger negative return thresholds, indicating greater vulnerability during adverse market conditions.

The VaR analysis highlights the importance of considering downside risk in addition to expected returns when evaluating investment opportunities.

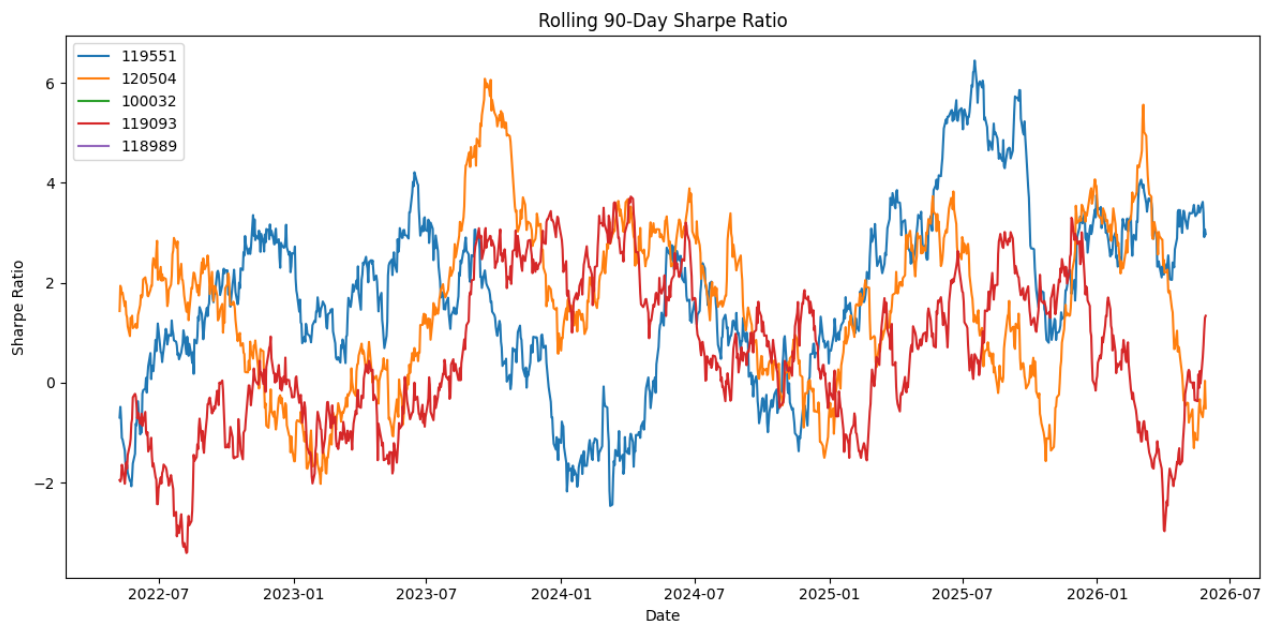
7.2 Conditional Value at Risk (CVaR)

While VaR estimates the threshold of potential losses, Conditional Value at Risk (CVaR) measures the average loss beyond the VaR threshold.

CVaR provides additional insight into tail risk and helps investors understand the severity of losses during extreme market events.

The analysis revealed that funds with higher VaR values also exhibited elevated CVaR levels, indicating greater exposure to severe market declines.

This metric is particularly valuable for risk-conscious investors and portfolio managers seeking to minimize extreme losses.



7.3 Rolling Sharpe Ratio Analysis

The Sharpe Ratio provides a snapshot of risk-adjusted performance over a fixed period. However, market conditions change continuously, making it important to evaluate performance over time.

To address this, Rolling 90-Day Sharpe Ratios were calculated for selected funds.

The analysis demonstrated that risk-adjusted performance fluctuated considerably throughout the study period. Certain funds maintained relatively stable Sharpe Ratios, while others experienced significant variations due to changing market conditions.

The rolling analysis provided a dynamic perspective of fund performance and highlighted periods of strong and weak risk-adjusted returns.

7.4 Investor Cohort Analysis

Investor behavior was analyzed by grouping investors according to the year of their first recorded transaction.

This approach enabled the identification of investment patterns across different investor cohorts.

The results revealed that:

- The 2024 cohort contributed approximately ₹349 crore in investments.
- The 2025 cohort contributed approximately ₹3 crore.
- Earlier cohorts generally accumulated larger investment balances.

The findings suggest that investors who enter the market earlier tend to accumulate significantly greater investment amounts over time due to sustained participation and continued contributions.

7.5 SIP Continuity Analysis

Systematic Investment Plans rely on consistent periodic contributions. Therefore, maintaining contribution discipline is critical for long-term wealth creation.

To evaluate SIP consistency, investors with six or more SIP transactions were analyzed.

The average gap between contributions was calculated, and investors were classified into two categories:

- Active Investors
- At-Risk Investors

Classification Results:

- Active Investors: 30
- At-Risk Investors: 1332

An investor was classified as At-Risk if the average gap between contributions exceeded 35 days.

The results indicate that a significant proportion of investors displayed irregular investment behavior. This finding highlights the importance of investor engagement and retention strategies within the mutual fund industry.

7.6 Portfolio Concentration Analysis (HHI)

Portfolio concentration was evaluated using the Herfindahl-Hirschman Index (HHI).

HHI is calculated as the sum of squared portfolio weights and serves as a measure of diversification.

Interpretation:

- Lower HHI = Better Diversification
- Higher HHI = Greater Concentration

The analysis revealed substantial differences in portfolio concentration across schemes.

Certain funds maintained highly diversified portfolios, while others exhibited greater dependence on a smaller number of holdings.

Concentrated portfolios may generate stronger returns when selected sectors perform well but are also exposed to higher stock-specific risk.

7.7 Fund Recommendation Engine

A simple recommendation engine was developed to assist investors in identifying suitable funds based on risk preferences.

The recommendation system used:

- Risk Category
- Sharpe Ratio
- Historical Performance

as the primary decision factors.

The engine categorizes investors into:

- Low Risk
- Moderate Risk
- High Risk

For Moderate Risk investors, the top recommendations included:

1. Mirae Asset Large Cap Fund
2. Kotak Flexicap Fund
3. SBI Bluechip Fund

These recommendations were selected based on superior risk-adjusted returns and strong overall performance metrics.

7.8 Business Value of Advanced Analytics

The advanced analytics framework significantly enhanced the practical value of the project.

Benefits include:

- Improved risk assessment
- Better investor profiling
- Enhanced portfolio analysis
- More informed investment decisions
- Identification of investor retention risks
- Simplified fund recommendation process

7.9 Key Findings from Advanced Analytics

The major findings from this stage were:

- Certain mid-cap and growth-oriented schemes exhibited higher downside risk.
 - Rolling Sharpe analysis revealed changing performance characteristics over time.
 - The 2024 investor cohort represented the largest investment segment.
 - More than 1300 investors were classified as At-Risk in SIP continuity analysis.
 - Portfolio concentration varied substantially across mutual fund schemes.
 - The recommendation engine successfully identified suitable funds based on investor risk profiles.

8. DASHBOARD DEVELOPMENT

One of the primary objectives of this project was to develop an interactive dashboard that could present analytical findings in a simple, accessible, and visually appealing manner. To achieve this objective, Streamlit was selected as the dashboard development framework due to its ease of use, flexibility, and seamless integration with Python-based analytical workflows.

The dashboard serves as a centralized platform that allows users to explore mutual fund performance, industry trends, risk metrics, and investor behavior without requiring direct interaction with the underlying datasets.

8.1 Dashboard Architecture

The dashboard was built using a modular architecture where each analytical component was organized into separate sections.

The overall workflow is as follows:

Raw Data Sources
↓
ETL Pipeline
↓
SQLite Database
↓
Analytics Modules
↓
Streamlit Dashboard

This architecture ensures that data processing and visualization remain separate, improving maintainability and scalability.

8.2 Dashboard Features

The dashboard provides the following capabilities:

- Industry Overview
- AUM Analysis
- SIP Trend Analysis
- Folio Growth Analysis
- Category Flow Analysis
- NAV Performance Analysis
- Fund Ranking Analysis
- Performance Analytics
- Risk Analytics

These modules allow users to explore both high-level industry trends and detailed fund-specific analytics.

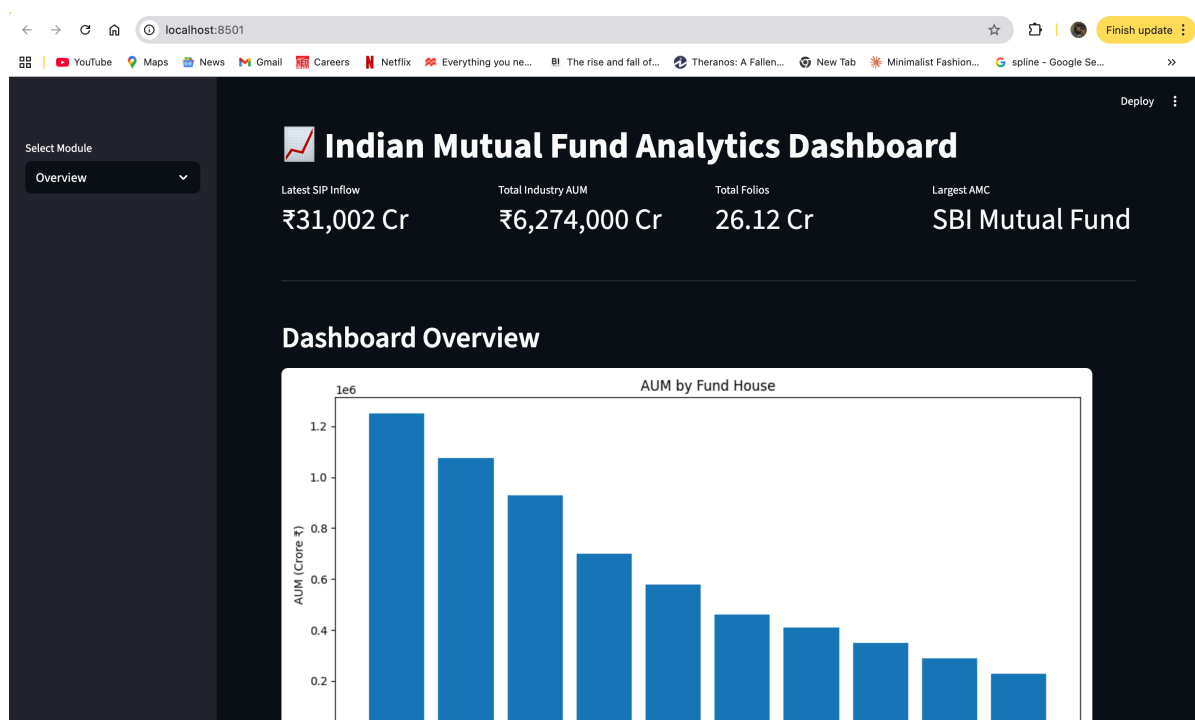
8.3 Overview Module

The Overview module presents key industry indicators through KPI cards and visualizations.

Important metrics displayed include:

- Latest SIP Inflow
- Total Industry AUM
- Total Investor Folios
- Largest Asset Management Company

This module provides users with a quick summary of the current state of the mutual fund industry.



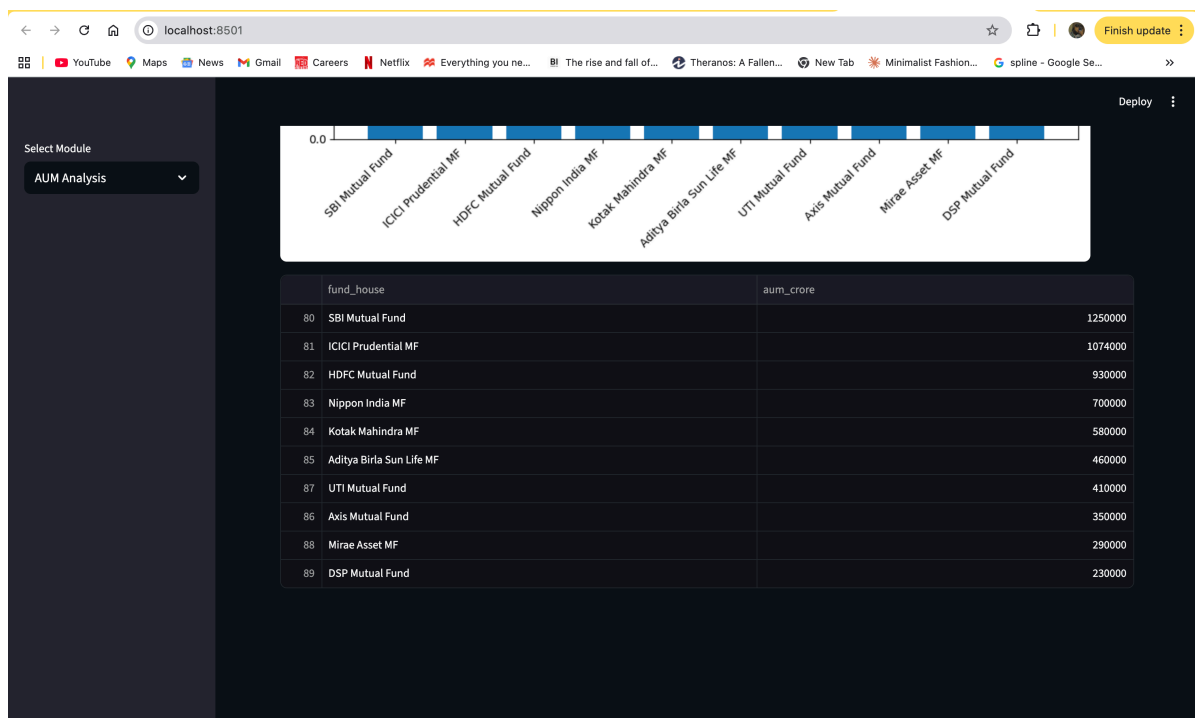
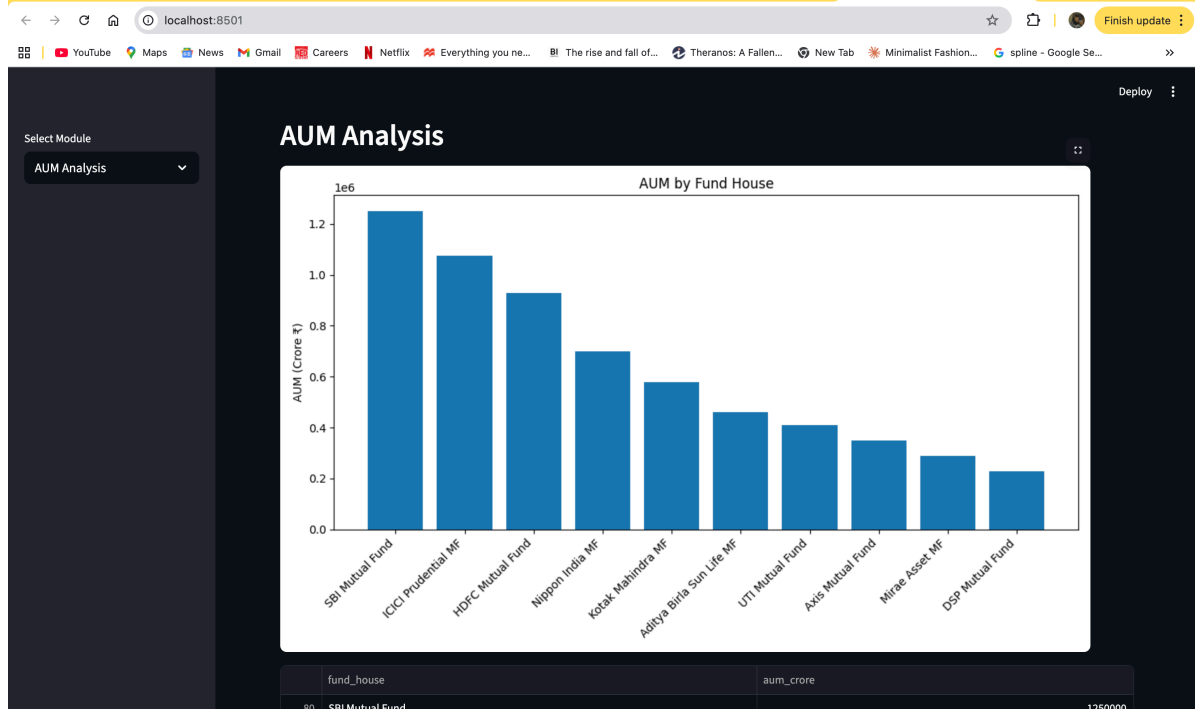
8.4 AUM Analysis Module

The AUM Analysis module enables users to evaluate the relative size and market presence of different fund houses.

The module includes:

- AUM visualization
- Fund house rankings
- Market share comparison

This view helps identify dominant industry players and understand industry concentration.



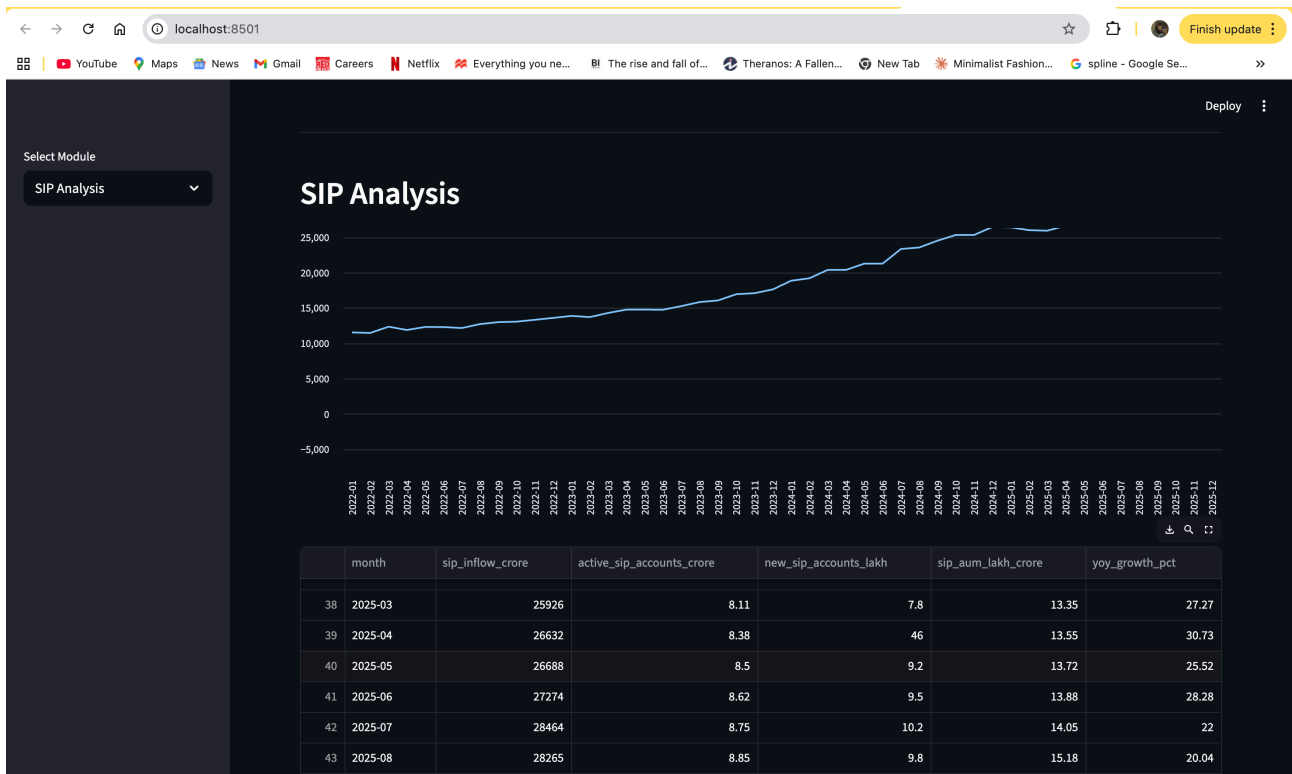
8.5 SIP Analysis Module

The SIP Analysis module focuses on industry-wide investment trends.

Key features include:

- SIP growth visualization
- Monthly inflow tracking
- Historical trend analysis

The module highlights the growing popularity of SIP-based investing among retail investors.



8.6 Fund Ranking Module

The Fund Ranking module displays the results of performance analytics and scorecard calculations.

Users can compare:

- CAGR Rankings
- Sharpe Ratio Rankings
- Alpha Rankings
- Composite Fund Scores

This module simplifies fund evaluation by presenting performance metrics in an organized format.

The screenshot displays the 'Fund Rankings' module. On the left, a sidebar shows 'Select Module' with 'Fund Rankings' selected. The main area features a table titled 'Fund Rankings' with 9 columns: amfi_code, cagr_1y, cagr_3y, scheme_name, fund_house, sharpe_ratio, alpha, beta, and max_drawdown. The table contains 10 rows of data for various mutual funds. Above the table, there are four large numbers: 148,002 CI, 10,214,000 CI, 20.12 CI, and SBI Mutual Fund.

	amfi_code	cagr_1y	cagr_3y	scheme_name	fund_house	sharpe_ratio	alpha	beta	max_drawdown
0	148567	0.2036	0.34	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	1.4483	0.2712	-0.0281	-0.1127
1	120505	0.296	0.3178	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	1.1801	0.293	-0.0174	-0.1819
2	120843	0.2666	0.2958	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	1.3067	0.2728	-0.0087	-0.1297
3	100033	0.5323	0.3244	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	1.0937	0.2723	-0.0112	-0.1622
4	120504	0.1306	0.3249	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	1.0265	0.2121	0.017	-0.1259
5	119094	0.2226	0.3511	Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	0.9982	0.26	-0.0599	-0.2096
6	119551	0.6044	0.3046	SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund	1.2083	0.2322	-0.056	-0.1501
7	148569	0.3975	0.2918	Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF	1.2349	0.2835	-0.0102	-0.164
8	101206	0.4792	0.2897	ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla Sun Life MF	1.0272	0.2139	0.0338	-0.1129
9	119598	0.8278	0.2667	SBI Small Cap Fund - Regular Plan - Growth	SBI Mutual Fund	0.9453	0.3011	0.0743	-0.2871

8.7 Performance Analytics Module

The Performance Analytics module provides detailed information regarding fund returns and risk-adjusted performance.

Metrics available include:

- Daily Returns
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

The module enables users to evaluate funds from multiple performance perspectives.

Select Module

Performance Analytics

Performance Analytics

Top CAGR Funds

	amfi_code	cagr_1y	cagr_3y	scheme_name	fund_house
0	119094	0.2226	0.3511	Axis Midcap Fund - Regular - Growth	Axis Mutual Fund
1	148567	0.2036	0.34	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF
2	120504	0.1306	0.3249	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF
3	100033	0.5323	0.3244	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund
4	120505	0.296	0.3178	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF
5	119551	0.6044	0.3046	SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund
6	120843	0.2666	0.2958	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF
7	148569	0.3975	0.2918	Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF
8	101206	0.4792	0.2897	ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla Sun Life MF
9	149324	0.6514	0.27	DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund

Top Sharpe Ratio Funds

	amfi_code	annual_return	annual_volatility	sharpe_ratio	scheme_name	fund_house
0	148567	0.2706	0.1419	1.4483	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF
1	120843	0.2726	0.1589	1.3067	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF
2	148569	0.2833	0.1767	1.2349	Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF

Select Module

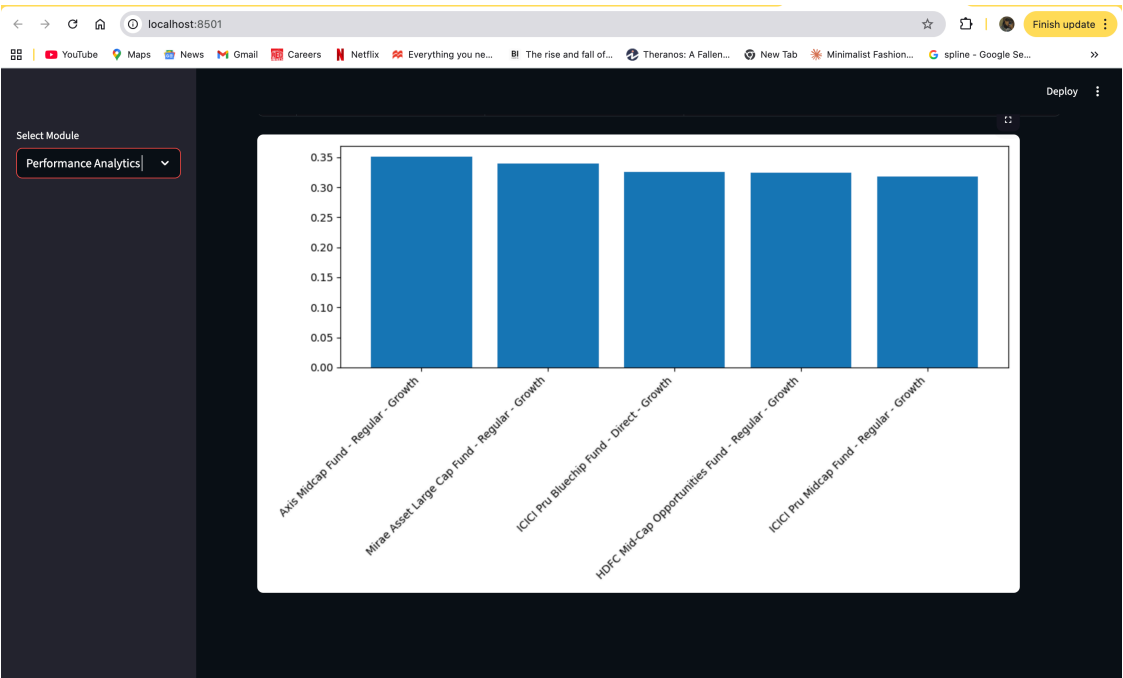
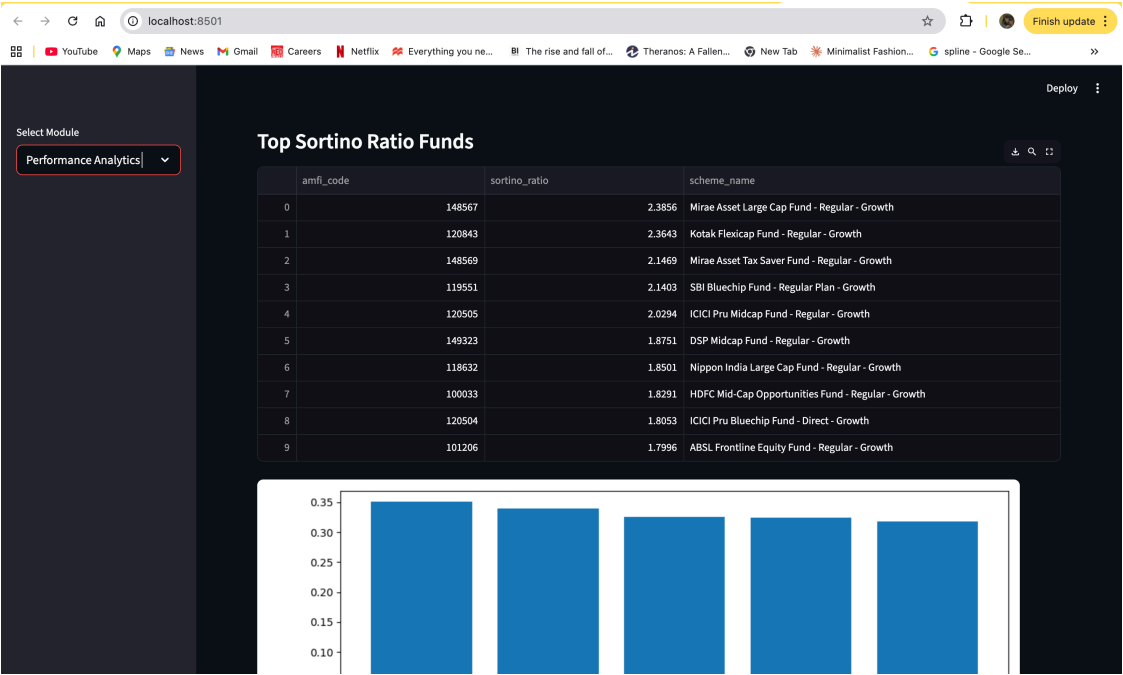
Performance Analytics

Top Sharpe Ratio Funds

	amfi_code	annual_return	annual_volatility	sharpe_ratio	scheme_name	fund_house
0	148567	0.2706	0.1419	1.4483	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF
1	120843	0.2726	0.1589	1.3067	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF
2	148569	0.2833	0.1767	1.2349	Mirae Asset Tax Saver Fund - Regular - Growth	Mirae Asset MF
3	119551	0.231	0.1374	1.2083	SBI Bluechip Fund - Regular Plan - Growth	SBI Mutual Fund
4	120505	0.2927	0.1929	1.1801	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF
5	149323	0.2659	0.1775	1.1321	DSP Midcap Fund - Regular - Growth	DSP Mutual Fund
6	100033	0.2721	0.1894	1.0937	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund
7	118632	0.218	0.1415	1.0817	Nippon India Large Cap Fund - Regular - Growth	Nippon India MF
8	101206	0.2146	0.1457	1.0272	ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla Sun Life MF
9	120504	0.2124	0.1436	1.0265	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF

Top Sortino Ratio Funds

	amfi_code	sortino_ratio	scheme_name
0	148567	2.3856	Mirae Asset Large Cap Fund - Regular - Growth
1	120843	2.3643	Kotak Flexicap Fund - Regular - Growth
2	148569	2.1469	Mirae Asset Tax Saver Fund - Regular - Growth
3	119551	2.1403	SBI Bluechip Fund - Regular Plan - Growth



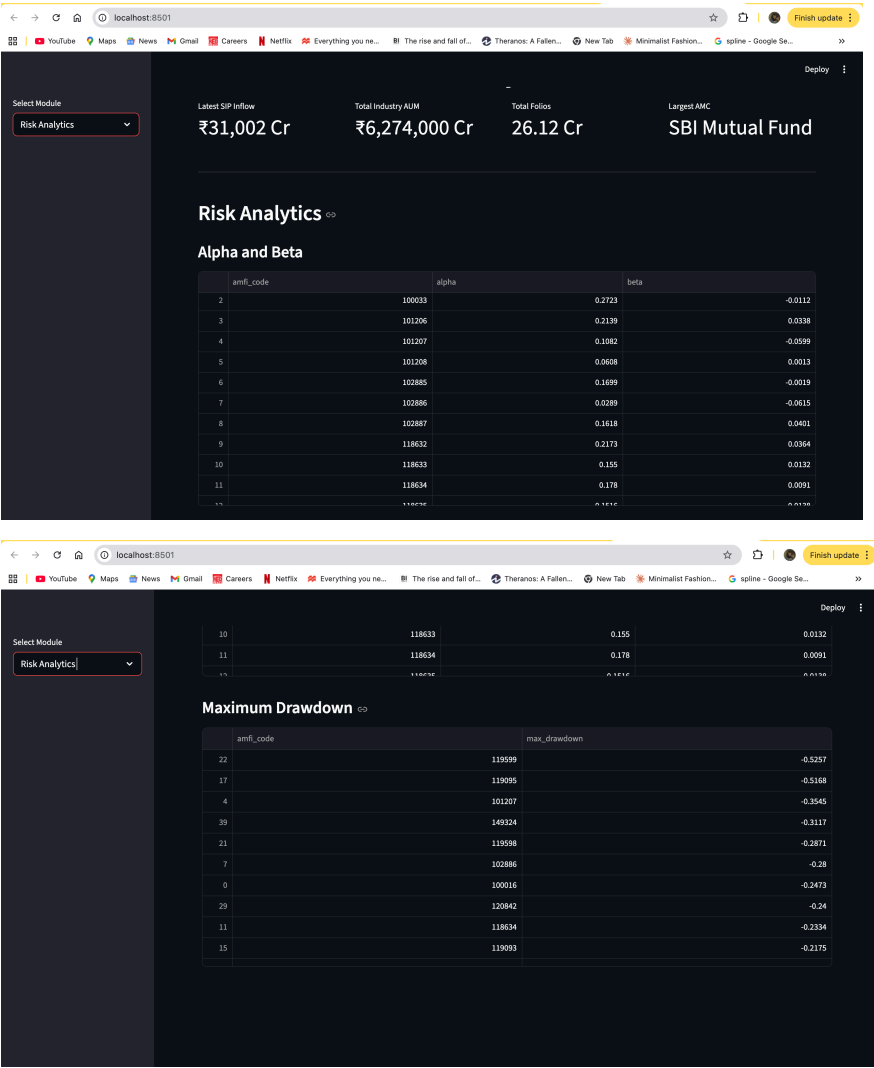
8.8 Risk Analytics Module

The Risk Analytics module focuses on downside risk and portfolio characteristics.

Metrics include:

- VaR
- CVaR
- Rolling Sharpe Ratio
- HHI Concentration Score

This module assists investors in understanding risk exposure and portfolio diversification.



8.9 Benefits of the Dashboard

The dashboard offers several practical benefits:

- Simplified data exploration
- Improved decision-making
- Real-time visualization of analytical outputs
- Enhanced user experience
- Centralized access to key metrics

The dashboard successfully transforms complex analytical outputs into an intuitive decision-support tool.

9.KEY FINDINGS

The project generated several important insights regarding mutual fund performance, investor participation, and portfolio risk.

The major findings are summarized below:

1. SBI Mutual Fund emerged as the largest asset management company with Assets Under Management exceeding ₹12,50,000 crore.
2. SIP inflows increased from approximately ₹11,517 crore to ₹31,002 crore during the study period, representing growth of more than 169%.
3. Total investor folios crossed 26 crore, indicating continued growth in retail participation.
4. Banking and Information Technology were the most represented sectors across mutual fund portfolios.
5. Mirae Asset Large Cap Fund achieved the highest Sharpe Ratio and demonstrated superior risk-adjusted performance.
6. Several mid-cap schemes generated strong CAGR values but also exhibited higher volatility.
7. Certain schemes recorded significantly higher Value at Risk levels, indicating elevated downside exposure.
8. The 2024 investor cohort contributed the highest investment volume among all analyzed cohorts.
9. More than 1,300 investors were classified as At-Risk due to irregular SIP contribution behavior.
10. Portfolio concentration varied substantially across schemes, highlighting differences in diversification strategies.
11. Risk-adjusted metrics provided more meaningful rankings than simple return comparisons.
12. The recommendation engine successfully identified suitable funds based on investor risk preferences.

10.Limitations

Although the project successfully achieved its objectives, several limitations should be acknowledged.

The analysis relied exclusively on historical data. As a result, the findings do not guarantee future fund performance.

Investor transaction data used within the project was synthetic and intended solely for educational purposes. Therefore, investor behavior findings should not be interpreted as actual industry statistics.

The project did not incorporate live market feeds or real-time NAV updates. All analyses were performed using static datasets.

Portfolio optimization techniques were not implemented, and the recommendation engine utilized a relatively simple ranking methodology.

Additionally, macroeconomic variables such as inflation, interest rates, and geopolitical events were not explicitly incorporated into the analytical framework.

Despite these limitations, the project provides a strong foundation for future development and enhancement.

11.RECOMMENDATIONS

Based on the findings obtained during the project, several recommendations are proposed.

First, future versions of the platform should integrate live NAV feeds and real-time market data to improve relevance and usability.

Second, machine learning models can be incorporated to forecast returns, volatility, and investor behavior.

Third, investor segmentation can be expanded using additional demographic and behavioral attributes.

Fourth, portfolio optimization techniques such as Modern Portfolio Theory can be implemented to support asset allocation decisions.

Fifth, the recommendation engine can be enhanced by incorporating multiple risk factors and investor preferences.

Finally, cloud deployment can improve accessibility and enable broader adoption of the platform.

12.FUTURE SCOPE

The project offers numerous opportunities for future enhancement.

Potential future developments include:

- Live NAV integration
- Real-time dashboard updates
- Machine learning-based forecasting
- Personalized investor recommendations
- Automated portfolio monitoring
- Portfolio optimization models
- Mobile dashboard applications
- Cloud-based deployment
- Integration with brokerage APIs
- Advanced risk management tools

These enhancements would transform the current analytics platform into a more comprehensive investment decision-support system.

13.CONCLUSION

The Mutual Fund Analytics Dashboard project successfully achieved its objective of developing an end-to-end analytics platform for mutual fund data.

The project integrated multiple datasets covering fund performance, industry growth, portfolio holdings, and investor transactions. Through the implementation of a structured ETL pipeline, robust database design, exploratory analysis, performance analytics, advanced risk modeling, and dashboard development, meaningful insights were generated from complex financial datasets.

The analysis highlighted important trends within the mutual fund industry, including significant SIP growth, expanding investor participation, variations in portfolio concentration, and differences in risk-adjusted performance across schemes.

The Streamlit dashboard successfully transformed analytical outputs into an interactive and user-friendly interface, enabling efficient exploration of key metrics and insights.

Overall, the project demonstrates how modern data analytics techniques can be applied to financial data to support better decision-making, improve transparency, and enhance understanding of investment performance. The resulting platform provides a strong foundation for future enhancements and real-world applications in the financial services domain.

REFERENCES

- 1.Association of Mutual Funds in India (AMFI)
- 2.Reserve Bank of India (RBI)
- 3.Pandas Official Documentation
- 4.NumPy Official Documentation
- 5..Streamlit Official Documentation
- 6.SQLite Official Documentation
- 7.Matplotlib Official Documentation
- 8.Financial Risk Management Literature
- 9.Mutual Fund Industry Reports and Publications
- 10.APPENDIX

Appendix A – Database Schema

Dimension Tables:

- dim_fund

Fact Tables:

- fact_nav
- fact_aum
- fact_sip
- fact_folio
- fact_category_flow
- fact_holdings
- fact_benchmark

Appendix B – Project Deliverables

- ETL Pipeline
- Data Cleaning Framework
- SQLite Database
- Exploratory Data Analysis
- Performance Analytics
- Advanced Analytics
- Recommendation Engine
- Streamlit Dashboard
- Final Report
- Presentation

Appendix C – Project Outcomes

The project successfully completed all planned objectives and deliverables. It demonstrates practical application of data engineering, data analysis, financial analytics, risk modeling, dashboard development, and software engineering concepts within a real-world financial analytics context.